Intro to Database & Big Query,

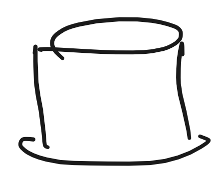
SQL → Structured Query Language.

Ledger. Pre - digital.

← data was stored in a book

Post digital.

Data stored in $\xrightarrow{DB}$



Where is the data stored?

What is DBMS.

What is schema?

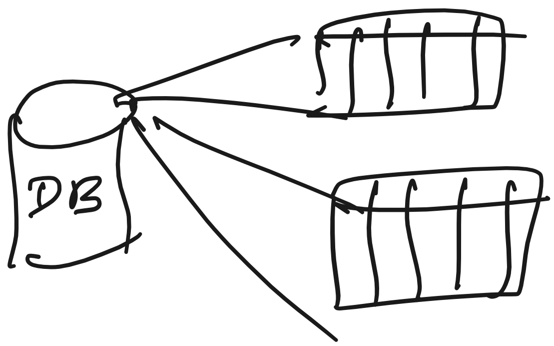
What are the boxes.

What are the lines,

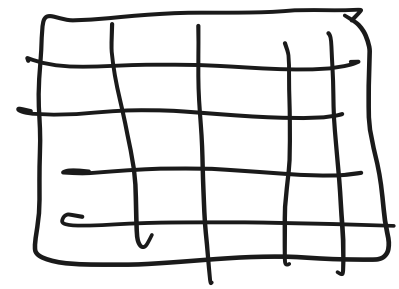
What is DBMS?

Data Base management system.

DB → collection of tables.



Tables are.



Management System.

Set of operation in managing or manipulating the DB.

CRUD → create, read, update, delete.
Search, insert, grant, revoke etc.

Date	Cus.id	Prod id	Sales
Jan 1	C1	P4	100
Jan 2	C3	P1	200
⋮	⋮	⋮	⋮

P4.

Relational DB

The tables in DB is also called relation.

Transac. table

Cus. table.

c id	Name

Trid	Pid	Cid	Cost
T1	P1	C1	100
T2	P2	C2	150
T3	P1	C1	40

DB.

C1	John
C2	Alan.
C3	Pete

Pid	Name
P1	Milk
P2	Egg.
P3	Butter.

Types of RDBMS.

- Oracle
- Sybase
- SQL lite.
- Postgres
- MySQL
- Redshift

Non-relational DB.

- MongoDB
- Cassendra.
- DynamoDB.

{key : value}.

{"name": Thanish,
"age": 34}.

{name : Alan
age : 30.
Country : USA},

Image / Video.

ER- diagram.   Blueprint of DB.

Cus_ info:

Name.
Age
Address
Email
Ph.no.
DOB.

Product:

Price
Category.
Brand
Sub-cat
Weight.

Customer purchase

P.id	v.id	M.date	cus.id	qty	C2C	time
Milk	V1	Jan 1st	C1	2	20	8:00AM
Butter	V2	Jan 1st	C1	4	80	9:30AM
⋮						

Data types.

108   "108"

- String → textual data.
- numeric → stores numeric info.
- Date time → stores date / time

String → Char(), varchar()

Char → fixed or the memory required is fixed

Char(3) → used to store columns with strings of 3 chars.

Country char(3) → accept only 3 characte

Country	
Ind	
AUS	
RUS	
CHN	
⋮	
IN	→ × will not allow
Japan	← × will not accept.

IN__

↑ filled with space.

varchar (5) → upto 5 characters and only allocate memory for the text length.

Country	
India	← 5 chars
US	← 2 chars

UK
AUS ← 3 chars.
⋮
Russia ← ✗ ← 6 chars.

Numeric: int64 → whole numbers → 4, 8, 3, 2
float64 → decimal → 4.2, 3.6, 4.9.

Date time → YYYY-mm-dd.

Keys:

cus_id	name	ph.no.	Age
1	Akon	9324	17
2	Akon		19
3	Bkon	2894	18
4	Ckon	5962	19
5	Dkon	2324	

Primary key: To identify each row.

- Should be unique
- Cannot contain Null.

- Every table should have a P.K.

Foreign key: P.K. of one table is foreign key in another table (if present)
F.K. can have dupliates also Null.

Transac table

Trid	Pid	Cid	Cost
T1	P1	C1	100
T2	P2	C2	150
T3	P1	C1	40

P.K.   F.K.   F.K.

Cus. table.

Cid	Name
C1	John
C2	Alan
C3	Pete

P.K.

Product

Pid	Name
P1	Milk
P2	Egg.
P3	Butter

P.K.

Doubt session:

cid.	Company	Name
C1	ABC	John
C2	XYZ	Alan
C2	ABC	Mike

Prod.id	Country	Sales
ip 11	India	10
ip 12	India	13
ip 11	US	20

Transac table

Trid	P.id	Cid	Cost
T1	P1	C1	100
T2	P2	C2	150
T3	P1	C1	40
T4	P3	Null	100
P.K.	F.K.	FK.	

cus.table.

Cid	Name
C1	John
C2	Alan
C3	Pete
P.K.	